

ÖMER GADALLA

Aerospace Engineer

Website: [omergadalla](https://omergadalla.github.io) || Email: omarnayer02@gmail.com || Phone: +90 552 796 03 00 || LinkedIn: [LinkedIn](#)

EDUCATION

Middle East Technical University (METU)

March 2021 – January 2026

B.Sc. Aerospace Engineering

EXPERIENCE

GALAXY ROCKET TEAM

December 2024-Present

Aerodynamics Engineer

TEKNOFEST Liquid Bipropellant Rocket Competition (A7) | 2026

- Defined aerodynamic drag coefficients across the full flight envelope using ANSYS Fluent external CFD, directly feeding results into vehicle flight performance models.
- Characterized combustion chamber and co-axial swirl injector flow behavior through CFD to validate nozzle and injector geometries and quantify performance losses.
- Contributed to system level trade studies balancing total impulse, apogee, and propellant mass fraction to meet competition design targets.

Propulsion Engineer

TEKNOFEST Liquid Bipropellant Rocket Engine Development Competition (B3) | 2025-2026

- Led all the analysis of a 100N LOX/Jet-A1 rocket engine, covering combustion chamber, nozzle, and propellant feed system from design to hot fire test.
- Modeled dynamic propellant feed system behavior in MATLAB Simscape, capturing transient flow response and validating subsystem interactions against design requirements.
- Resolved combustion and nozzle flow characteristics through ANSYS Fluent CFD, identifying performance margins and informing design iterations.
- Verified structural integrity of the combustion chamber under 3,000 Kelvin and high chamber pressure through coupled thermo-structural FEA.
- Quantified co-axial swirl injector atomization characteristics including spray cone angle, droplet breakup, and liquid film behavior using VOF and DPM multiphase CFD.
- Validated VOF/DPM simulation predictions against cold flow injector test data, confirming model accuracy.
- Supported component level valve testing including functional, leak, and pressure verification up to 1.5× MEOP.
- **Achieved Finalist ranking among competing national teams.**

CFD & Structures Analysis Team Lead

TEKNOFEST High Altitude Rocket Competition (A3) | 2025

- Led the CFD and structural analysis team for a 2-meter rocket achieving Mach 1.4 and 15,000 ft apogee.

MTS Defense Industry

July 2025-September 2025

R&D Engineer – Intern

- Independently developed the conceptual design of a 20N green bipropellant thruster for 10-100 kg small satellites.
- Communicated system architecture and development risks directly to system engineers and company investors.

Skills

CFD: Ansys FLUENT (DesignModeler, SpaceClaim, Meshing), OpenFoam (Paraview), Multiphase (VOF, DPM), Reacting Flow (Species Transport, EDC), External Aerodynamics, XFOIL, DATCOM

CAD: Inventor, SolidWorks, Fusion 360

Structural: Ansys, Nastran, Thermo-Structural

Programming: Matlab, Simulink/Simscape, Nasa CEA

Languages: English (Fluent), Turkish (Fluent), Arabic (Fluent)

For a summary of my engineering projects I have worked on and my skills, check out my online portfolio at:

<https://omergadalla.github.io/portfolio/>